A Problem with Presidents: Lifespan Data Analysis and Organization
CTS-IT Application Interview Project

Emily Forestier
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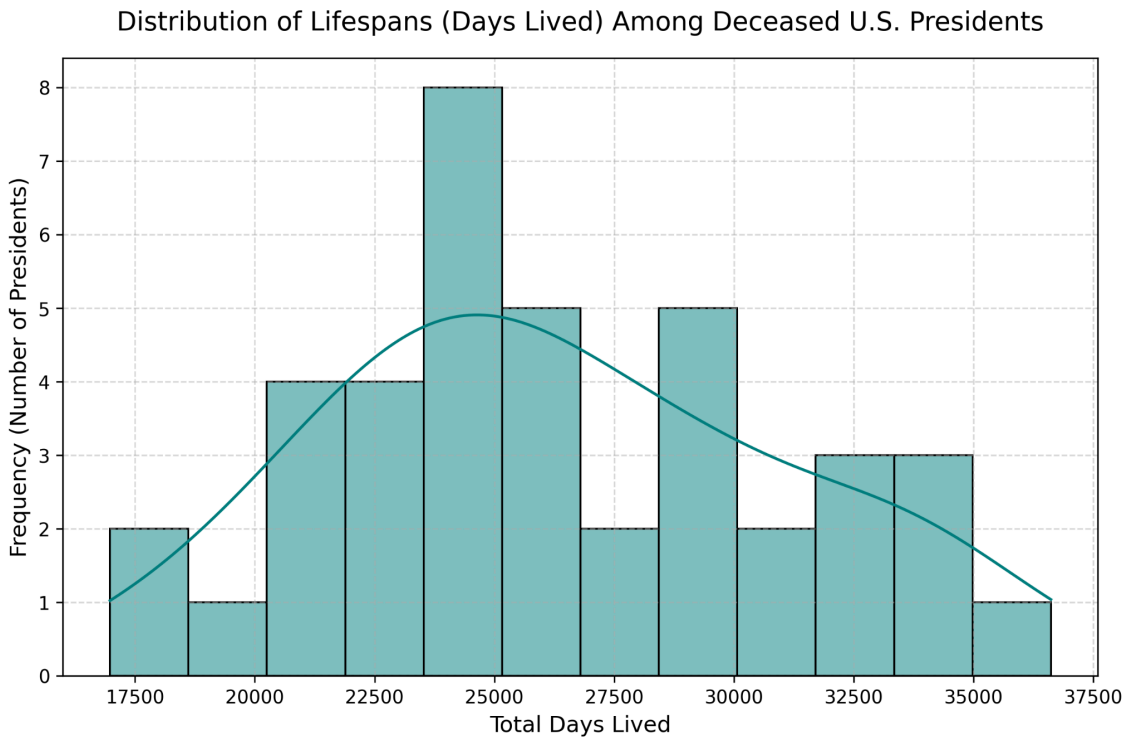
Abstract

In this project, by organizing and analyzing each deceased president's longevity statistics, it is concluded that President George Bush lived the longest and John F. Kennedy lived the shortest. The histogram created with Matplotlib shows a bell-shaped distribution, with the largest bin indicating that 6 deceased presidents lived ~24,000 days (around 65 years). The extremes of the statistics also highlight that 2 presidents lived ~17,500 days (around 46 years) and four lived 32500+ days (around 90 years).

Methodology

After converting the entire CSV file into a Pandas DataFrame, individual columns, such as "bday" and "dod," were created to enable large-scale calculations efficiently. This was done by converting the file's strings into datetime objects. Based on the information, the deceased presidents were organized into their own table since they had a valid lifespan, making statistical calculations possible. With the new table, the number of days, years, and months lived were calculated to organize the presidents into lists of longest- and shortest-lived. The days were used as the baseline metric for comparing overall longevity of each president's life, and basic math functions provided the final numbers.

Data



Graph 1: Histogram Statistical Summary of Deceased U.S. President Lifespans

PRESIDENT	year_of_birth	lived_years	lived_days
George Bush	1924	94.466804	34504
Gerald Ford	1913	93.451061	34133
Ronald Reagan	1911	93.327858	34088
John Adams	1735	90.67488	33119
Herbert Hoover	1874	90.193018	32943
Harry S. Truman	1884	88.632444	32373
James Madison	1751	85.284052	31150
Thomas Jefferson	1743	83.22245	30397
Richard Nixon	1913	81.281314	29688
John Quincy Adams	1767	80.618754	29446

Table 1: Top 10 Longest-Lived Presidents alongside their year of birth, the number of days they lived, and the number of days they lived

PRESIDENT	year_of_birth	lived_years	lived_days
John F. Kennedy	1917	46.483231	16978
James A. Garfield	1831	49.83436	18202
James K. Polk	1795	53.615332	19583
Abraham Lincoln	1809	56.169747	20516
Chester Arthur	1829	57.119781	20863
Warren G. Harding	1865	57.744011	21091

William McKinley	1843	58.622861	21412
Theodore Roosevelt	1858	60.19165	21985
Calvin Coolidge	1872	60.503765	22099
Franklin Roosevelt	1882	63.195072	23082

Table 2: Top 10 Shortest-Lived Presidents alongside their year of birth, the number of days they lived, and the number of days they lived

Statistic	Value
Mean	26097.17949
Median	25673
Min	16978
Max	34504
Standard Deviation	4596.072413

Table 3: Statistical Summary of All Deceased Presidents